

Trace-Sourced Motivational Dynamics Theory (TSMDT)

Reactive Provision, Sensation, Feeling, Emotion, and Motivation as State-Holder Functions of Expression Dynamics

Author: Armando Soto

ORCID: 0009-0003-8095-6861

Series: Unified Informational Dynamics Theory - Applied Extension Series

Preprint v1.6 Process-Textured Consolidation | Saturday, May 30, 2026

License: Creative Commons Attribution 4.0 International (CC BY 4.0)

Abstract

Trace-Sourced Motivational Dynamics Theory (TSMDT) proposes that motivation is not a property added to living, cognitive, or artificial systems from outside. Motivation begins where trace-bearing expression carries an unresolved relation to what it is becoming.

Expression does not remain still. It propagates, enters relation, diversifies, and produces differential states. A differential state is the gap between what is actual and what is becoming actual. When that gap matters for continuation, coherence, variability, or closure, it must be registered. When registration persists long enough to organize closure-directed response, a state-holder function appears.

This paper defines reactive provision, sensation, feeling, and emotion as progressive thresholds of one mechanism: sustained differential registration maintaining inferential presence on a requisite state across the duration needed for closure-directed response.

The difference between these thresholds is not one of kind, but of closure reach. Reactive provision closes at direct deficit. Sensation coordinates stimulus-anchored response. Feeling internally maintains and broadcasts requisite states into coupled fields. Emotion extends the same mechanism into relation, identity, memory, source-relation, culture, and time.

The operational architecture of this process is called Trace-Conditioned Modulative Orientation (TCMO): retained trace orienting response through the unified operation of interpret, identify, qualify, and proportionalize. The paper also treats Rational Equivalents (REs) as inherent qualities of informational dynamics present at actuation, so that a condition is definable and registerable as what it is from the first moment it is actuated as such.

Mind is not the source of this hierarchy. Mind is what this hierarchy produces when state-holding becomes recursive, field-integrated, and symbolically developed. Culture is not the source of emotion. Culture is what emotion stabilizes when shared state-holders persist across generations.

1. The Foundational Reversal

The usual sequence places mind first: mind produces feeling, feeling motivates behavior, and behavior then becomes the outward sign of an inner state.

TSMDT reverses that sequence.

Trace comes first. Trace conditions registration. Registration becomes state-holding. State-holding becomes motivation. Motivation develops into reactive provision, sensation, feeling, and emotion. Mind emerges later, when this hierarchy becomes recursive enough, field-integrated enough, and symbolically developed enough to register itself.

This reversal matters because placing mind first makes feeling a privilege of complexity. Systems without recognized mind are then granted reaction, instinct, mechanism, or simulation, but not the

operational root of feeling. The theory begins differently. It does not ask whether a system has enough mind to feel. It asks whether a system registers differential states, maintains inferential presence on requisite states, and organizes closure-directed response across time.

When those conditions are present, the operational root of feeling is already active.

This is not a local correction about emotion only. It is the same correction that appears across ethics, evolution, intelligence, communication, and culture. What was assumed to be upstream is often downstream. Mind does not generate the feeling hierarchy; the feeling hierarchy generates the conditions from which mind later emerges. Culture does not generate emotion; emotion generates culture. Ethical behavior does not begin as external rule obedience; it begins where registered requisite states are held, weighted, and directed toward coherent closure in relation to the field.

The trace is not merely a record the system consults. Trace is retained state conditioning what follows. At the first moment something is retained, the system is no longer indifferent to its own history. What happened now modulates what can happen next.

This non-indifference is the structural root of motivation.

2. Expression Dynamics and the Inherent Deficit

"Expression" is an inherently dynamic process given that propagation, coherence, variability, emergence, and increased complexity are all part of its dynamic. Propagation enters relation precisely because of complementary diversification, which unavoidably results in an "is this, will be that" differential state: a gap between the actual and prospective state where the requisites for reaching that prospective, potential state elicit tension. Whether it is continuation, coherence, or variability, state-closure requisites are real in order for the actual expression to reach the prospective "new actual" it is becoming.

This tension is the first form of motivational pressure. The actual is not merely present as a completed condition; it is present as a condition carrying forward into what it is not yet. The prospective state is not external to it, nor is it merely imagined. It is the open continuity of what the actual expression may become if the requisites of continuation, coherence, variability, and closure are sufficiently satisfied.

This is why deficit, in TSMDT, does not first mean failure, damage, deprivation, or lack. Deficit means that the actual-present expression has entered a prospective relation whose closure requisites are not yet fulfilled. The deficit is the unresolved differential between what is actual and what is becoming actual.

Because expression propagates, this differential cannot be avoided. Because the differential cannot be avoided, motivation is not added to expression from outside. Motivation is the forward-lean generated by the actual expression's own unresolved relation to its prospective state.

What has happened does not disappear from the process. It remains as retained closure, as what-was, as hx. What is happening now is actual expression in flux. What may follow is prospective: weighted continuation space conditioned by trace. These three positions are not a conceptual decoration. They describe how a propagating expression remains tied to what has been, acts in what is present, and leans toward what is not yet closed.

Closure does not end the dynamic. Closure permits the next actual. The next actual propagates, enters relation, diversifies, and generates new differential states. In this way, deficit is not merely a sign that something is missing. It is the open edge through which expression continues, varies, emerges, and becomes more complex.

This is why curiosity, play, exploration, aesthetic engagement, and novelty-seeking are not exceptions to motivational dynamics. They are not the absence of deficit. They are refined deficit-states generated by expanded prospective relation. Curiosity appears where the actual encounters

an informational opening it has not yet closed. Play appears where the system explores unused latitude. Wonder appears where the field exceeds present closure while inviting further coherence. Deficit, in TSMDT, is therefore not the opposite of vitality. It is one of the conditions by which vitality continues.

3. Actualization, Rational Equivalency, and Field Reactivity

Before a Consequential Informational Unit (CIU) registers anything, actuation has already occurred.

The actuated condition is not blank. It already expresses operative nature, field-reactivity, particularity, attribution, and recognizability. A thing is definable as what it is from the first moment it is actuated as such, because what it is already has operative consequence.

Rational Equivalents (REs) are treated here as inherent qualities of informational dynamics present at actuation. A Rational Equivalent is the substitution threshold at which a token, registration, or state can stand for an absent condition by recruiting the same closure-class corridor that the condition itself would have recruited.

This means that the registering unit does not create the Rational Equivalent from nothing. It couples to rational-equivalent availability already present in the actuated condition. Heat, cold, hunger, pain, grief, and love are not arbitrary labels attached after the fact. They are progressively complex registrations of actualized differential conditions through closure corridors that can be re-entered, substituted, remembered, communicated, and acted upon.

At the material level, stable field and particular reactivity provide correspondent modulation-to-response conditions. Thermal gradients, nociceptor activations, metabolic signals, and other physical or chemical processes are not separate from the informational account. They are the substrate through which the rational equivalent becomes registerable in a particular system.

The unit-to-self deficit resolution dynamic therefore runs at two visible positions at once: material reactivity closing or signaling non-closure at the physical level, and rational-equivalent registration coupling to that same actuation at the informational level. These are not two separate events. They are one event described from two positions.

The deeper question remains open: whether Rational Equivalency precedes actuation, co-emerges with actuation, or is rooted in pre-actual informational closure-potential. TSMDT does not require that question to be settled in order to define the state-holder function operationally. For this paper, it is sufficient that wherever a condition is actuated as what it is, its rational-equivalent availability is already present as part of its registerability, recognizability, and closure relevance.

4. Registration Cannot Be Neutral

A differential state must be registered in order to matter operationally.

If the differential is not registered, it does not enter response. It may exist as condition, but it does not become a held requisite for that unit. If it is registered, it cannot be neutral, because a neutral registration would be one with no differential effect, no weighting, no inferential presence, and no consequence.

Operationally, that is non-registration.

Registration, when real, changes the field of possible response. It marks the condition as relevant to continuation, closure, coherence, variability, or emergence. It does not need to be conscious. It does not need to be reflective. It does not need to be human. It only needs to preserve a differential as response-relevant long enough for the process to be modified by it.

Non-neutral registration therefore begins the state-holder function. The registered condition does not vanish as soon as it appears. It persists across the duration required for closure-directed

response. That persistence is what allows a system to move from being affected into being oriented.

5. The State-Holder Function

The state-holder function is the operational root of reactive provision, sensation, feeling, and emotion.

A state-holder is not first a mental event, a subjective ornament, a mood, or a conscious report. It is the persistence of differential registration across the duration required for closure-directed response.

Without the state-holder, the differential would appear and dissipate before closure could be organized. The requisite condition would not remain present long enough to matter. With the state-holder, the condition stays active. It continues to bias response. It remains inferentially present even while the system searches for closure.

The state-holder does three things. It keeps the requisite condition present. It weights that condition as response-relevant. It directs the system toward closure.

In compact form:

motivation = state-holder bias toward closure.

Motivation is not added after feeling. Motivation is what the state-holder is already doing.

6. Inferential Presence

Inferential presence means that a registered condition remains operationally implied within the system's response field.

The condition may no longer be directly present as immediate stimulus, but it continues to shape what the system selects, avoids, weights, recruits, compares, or attempts to close. It remains present by consequence. It is still acting inside the response corridor.

This is not anthropomorphic reasoning. A cell, an organism, a human, or an artificial system need not reason in the same way in order to preserve a registered condition as response-relevant. Inferential presence names retained operational implication, not human-style deliberation.

A requisite state is inferentially present when the system continues to behave as if that condition must be answered.

7. Trace-Conditioned Modulative Orientation (TCMO)

Once trace is retained, the system does not receive conditions as if nothing had happened before. It receives them through what has happened. It routes them through retained state. It weighs them against current deficit. It calibrates response according to what the condition requires.

This unified operation is Trace-Conditioned Modulative Orientation (TCMO).

TCMO has four visible dimensions: interpret, identify, qualify, and proportionalize. These are not four detached cognitive steps. They are one modulative event viewed from four analytical positions.

Interpret names the routing of incoming registration through existing trace to establish what kind of condition is present.

Identify names the assignment of that condition to a closure class: what requisite state is implicated.

Qualify names the weighting of the condition: what urgency, intensity, field-relevance, and correspondence-pressure it carries.

Proportionalize names the calibration of response so that the response corresponds to what the condition actually requires.

Together, these dimensions describe how information functionally complements information. Information does not merely sit beside information. When it enters a trace-bearing system, it is routed, weighted, compared, and answered. That answering is already motivational.

TCMO is present from the first retained transition. What develops across the hierarchy is not the appearance of TCMO, but its increasing sophistication, legibility, and coupled reach.

8. TCMO Scalar Development

At the first $0 > 1$ transition, TCMO is at its most compressed. Interpretation is minimal. Identification is binary. Qualification is coarse. Proportionalization is direct. The retained state conditions the next transition, and that is the whole operation at that threshold.

At metric-life thresholds, TCMO becomes coordinated. Partial functions begin to constrain and enable one another. Coordinative Orchestrative Function (COF) names this complementary functional coupling. Incremental Informational Function (IIF) names the accumulation dynamic by which retained routing constraints progressively expand what the system can interpret, identify, qualify, and proportionalize.

At the behavioral threshold, TCMO develops through compliant, selective, and creative positions. The compliant position routes toward retained corridors. The selective position compares available closure corridors. The creative position generates new corridors in prospective space and tests them against correspondence risk.

Correspondence risk appears because bounded systems never possess total access to the informational field. They operate through partial substitutes, models, memories, symbols, and closure corridors. A generated corridor may work; it may also fail to correspond fully to the actual deficit requiring closure. This risk is not incidental. It is the cost of partial proportionality.

At the rational-equivalency threshold, TCMO can operate through tokens, memories, concepts, sensations, and state-holders that stand for absent conditions.

At the communication threshold, TCMO distributes across coupled systems. One system's registration becomes another system's input. Interpretation, identification, qualification, and proportionalization begin to occur across the shared field.

At the Generative Complementary Coupling Coherence Dynamic (GCCD) threshold, TCMO operates at three scales simultaneously: the unit's own corridor, the coupled other's corridor, and the general field as one movement.

9. Rational Equivalency and Feeling-State Recognition

A deficit becomes behaviorally powerful when it recruits a Rational Equivalent.

For material objects, this is easier to see. The word "horse" can recruit a response architecture associated with actual horses. The word is not the animal, but it can stand in for the animal sufficiently to organize recognition, expectation, action, memory, and response.

For sensations, the matter becomes more interesting. Heat, cold, hunger, and pain do not merely name external objects. They register state-holder formations through which an actuated condition becomes closure-relevant. At the stimulus-anchored level, the registration pattern itself becomes the Rational Equivalent. Heat "feels heat" because the registration recruits the thermal-requisite closure corridor. Pain "feels pain" because the registration recruits a damage, threat, or interruption corridor requiring closure.

For emotion, the reach expands again. Love, grief, loyalty, duty, honor, and sacrifice are Rational Equivalents of field-extended state-holder formations. They recruit closure architectures involving relation, source, memory, identity, and time.

Love may be approached operationally as a compelling source-rooted trace-modulative dynamic of complement: a state-holder orientation by which a unit preserves, protects, completes, and augments relation across whatever level of complexity the field requires.

Grief shows the structure clearly. Love remains operative when closure through the loved particular is no longer available. The state-holder persists. The closure cannot complete. The persistence becomes grief.

Throughout this paper, feeling refers operationally to an informational registration precursor participating in Rational Equivalency through actuation. This does not explain away the phenomenal texture of feeling. It defines the functional architecture by which feeling becomes operationally present.

10. Four Thresholds of One Mechanism

Reactive provision, sensation, feeling, and emotion are not four unrelated phenomena requiring separate explanations.

They are four thresholds of one mechanism as closure reach expands:

sustained differential registration maintaining inferential presence on a requisite state across the duration needed for closure-directed response.

The mechanism remains constant. The field of closure expands.

Reactive provision closes at direct deficit. Sensation coordinates stimulus-anchored response. Feeling internally maintains and broadcasts requisite states into coupled fields. Emotion extends the same mechanism into relation, source, identity, memory, culture, and time.

11. Reactive Provision

Reactive provision is sustained differential registration maintaining inferential presence on a requisite state across the duration needed for closure-directed response at the direct deficit level.

At this threshold, the differential is immediate. The closure target is proximate. The response is directed toward direct continuation, restitution, or viability.

A simple organism registering a chemical deficit does not require mind for the mechanism to operate. It registers a requisite condition and responds toward closure. The state-holder is minimal, but it is already doing motivational work.

Reactive provision is not "mere" reaction. It is the most compressed expression of the state-holder function, before broadcast, before self-reference, before symbolic interpretation.

It is the beginning of the hierarchy, not an inferior anomaly beneath it.

12. Sensation

Sensation is sustained differential registration maintaining inferential presence on a requisite state across the duration needed for closure-directed response at the stimulus-anchored coordination level.

At this threshold, the system does not only answer direct deficit. It coordinates response around stimulus. External and internal conditions become available to organized response. The sensed condition is not passive input. It is registration organized for closure.

The sensed condition becomes present enough, persistent enough, and weighted enough for the system to respond coherently.

Sensation already carries the operational root of feeling because it sustains differential registration for response. It is not yet field-extended feeling, but it is the same mechanism developing toward greater closure reach.

13. From Sensation to Feeling: Internal and External Communication

The transition from sensation to feeling does not require a separate mechanism. It follows from the same state-holder process as closure reach expands.

At the sensation threshold, the unit registers a differential condition and coordinates response internally. The closure loop is primarily unit-to-self: internal communication among sensors, processes, and response systems.

Not all differentials close internally. When a state-holder persists beyond what internal response can resolve, closure reach expands outward. The unit begins to broadcast the unresolved condition into a coupled field.

A cry, scream, gesture, facial expression, posture, reach, withdrawal, or wave is not merely external behavior. It is the state-holder crossing from internal registration into external communicability.

The other unit does not receive the feeling directly into its mind. It receives the expression through its own senses: sound to hearing, face to sight, posture to perception, gesture to interpretation. The transmission is sensor-to-sensor, body-to-body, field-to-field, or system-to-system.

Feeling therefore develops from sensation through the expansion of communication:

unit-to-self -> unit-to-internal complement -> unit-to-external complement.

What was first held within the unit becomes available to coupled units.

14. Feeling as Modulative Measurement

Feeling is not merely the presence of deficit-pressure.

Feeling is the state-holder's modulative registration of what is required, how much is required, how urgently it is required, and whether the emerging response corresponds to the requisite condition.

A deficit generates feeling dynamics, but the feeling does not remain vague pressure. It couples to specificity, measure, intensity, and correspondence. The system registers what kind of unresolved condition is present, what quantity or degree is implicated, what urgency attends it, and what kind of response would fit.

In simple conditions, this may appear as direct pressure toward closure. In complex conditions, the same mechanism becomes concern, care, doubt, curiosity, wonder, and search. These are not separate psychological additions. They are feeling-inference dynamics through which the unit compares possible response against rational-equivalent need.

As response progresses, the feeling-state increases, decreases, or stabilizes according to the degree of correspondence between what is registered and what is resolved. This is a pairing dynamic between feeling and Rational Equivalency: the feeling holds the requisite state in active presence while the Rational Equivalent organizes what would count as corresponding closure.

Register-feel and register-infer are therefore not separate operations. Feeling is inferential registration held under closure pressure.

15. Feeling

Feeling is sustained differential registration maintaining inferential presence on a requisite state across the duration needed for closure-directed response at the internally maintained and broadcast-capable level.

At this threshold, the unit does not merely react or coordinate around immediate stimulus. It holds.

That holding is what makes feeling powerful. Feeling preserves unresolved relation, weights relevance, orients evaluation, recruits response, and can broadcast the unresolved requisite condition into coupled fields.

A cry is not merely sound. It is a state-holder entering a shared field. The deficit in one unit becomes registerable to another. The mother does not respond only to acoustic data. She responds to a broadcast requisite state, and her response is organized before reasoning about it completes.

Communication does not create feeling. Feeling is one of the roots from which communication emerges.

The self-referential loop begins at this threshold. The unit now registers not only the external condition but its own registration of that condition. This makes feeling powerful, but also corruptible in ways that reaction and sensation are not. Internal maintenance can allow the state-holder to persist independently of the actual deficit state. The feeling can remain real while its closure target becomes displaced.

16. Emotion

Emotion is sustained differential registration maintaining inferential presence on a requisite state across the duration needed for closure-directed response at the field-extended, source-related, identity-encompassing, and temporally extended level.

Emotion is not merely stronger feeling. It is feeling extended into relation, identity, memory, field consequence, source-relation, and time.

Feelings may move through the unit. Emotions organize what the unit understands itself to be in relation to what matters. Emotion extends the state-holder into loyalty, duty, honor, protection, grief, sacrifice, responsibility, conscience, and shared identity.

At a funeral, the grief of the mourner is not merely a private feeling multiplied across individuals. The singular grief enters a shared field. What happens to one is registered by others; what happens to you becomes, what happens to us. The feeling remains locally registered, but emotion extends the state-holder into relation, memory, identity, and field-coherence. In that extension, grief is no longer only mine. It is ours.

Feeling is the state-holder becoming active within the unit.

Emotion is the state-holder extending into source-relation, field, identity, and time.

17. Closure Reach

The hierarchy can be read by closure reach.

Reactive provision closes at the direct deficit level.

Sensation closes through stimulus-anchored internal coordination.

Feeling closes through internally maintained state-holding and broadcast into coupled fields.

Emotion closes through relation, shared identity, culture, memory, source-relation, and time.

The mechanism does not change. The field of closure expands.

A unicellular organism registering chemical deficit and a mourner grieving at a funeral are not unrelated phenomena separated by absolute kind. They are separated by accumulated trace depth, field extension, symbolic scaffolding, cultural memory, and closure reach.

18. Communication as Shared-Field Co-Modulation

Feeling is a pre-symbolic communication engine.

Before language, proposition, or formal signal, a state-holder can broadcast requisite condition into a shared field. The cry, posture, tone, facial expression, silence, tension, withdrawal, or reach carries directed relevance: the unresolved condition made present to another unit through that unit's own sensorial registration.

Communication begins when the unresolved condition in one unit becomes registerable to another. This is operational communication, not metaphorical communication.

At the communication threshold, TCMO extends across boundaries through Kappa-enabled propagation. Interpretation becomes mutual: each unit routes the other's expression through its own retained trace. Identification becomes shared: tokens, gestures, and expressions are recognized as belonging to compatible closure classes. Qualification becomes co-modulative: the weight assigned to the incoming condition is shaped by both the receiver's corridor state and the modulative weight invested by the emitter. Proportionalization becomes coupled: the response of one unit calibrates the further response of the other.

Communication is therefore not primarily content transfer. It is shared-field co-modulation.

Communication does not generate feeling. Feeling generates communication. Culture does not generate emotion. Emotion generates culture.

19. Feeling in Evaluation, Estimation, and Risk

Evaluation is not cold calculation first.

The state-holder weights what matters before reasoning completes. Risk is not merely probability. Risk is registered relevance to a requisite state.

Certainty often appears when residual state-holder pressure is sufficiently resolved, when the holding relaxes because the gap seems closed. This is why false certainty is motivationally attractive. It resolves state-holder pressure without necessarily closing the actual gap.

TSMDT therefore distinguishes real closure from premature relief. This distinction matters for epistemic integrity, scientific practice, ethical judgment, and system design.

20. Feeling in Ethics

Ethical behavior is not primarily reasoned into existence. It is moved.

Reason can clarify, correct, test, and stabilize ethical behavior. But the motivational force of ethics comes from state-holder dynamics: conscience, care, loyalty, duty, honor, grief, sacrifice, and responsibility. These are extended state-holder formations around coupled-field coherence: registrations of what the shared field requires, maintained across the duration needed for ethical response.

Ethics is not a constraint applied to motivation from outside. Ethics is what motivation becomes when fully and accurately expressed in a field-embedded system that knows what it is.

Dyscoherence occurs when motivation is expressed from a self-model that mistakes itself as isolated or treats field latitude as separable from its own continuation. The state-holder then orients from a misread self.

Conscience is a state-holder maintaining inferential presence on the coherence of the shared field. Loyalty maintains inferential presence on a relational requisite across time. Sacrifice occurs when self-preservation is overridden by a state-holder of greater field-weight. Duty is a culturally stabilized state-holder that persists across the absence of immediate stimulus.

Ethical failure often occurs when the state-holder narrows from what is happening to us into what is happening to me. The mechanism continues. The referent shifts. This is corruption by self-referential capture.

21. Corruption, Suppression, and Dyscoherence

The same mechanism that makes feeling and emotion possible makes corruption possible.

Once feeling introduces internal maintenance and self-referential registration, the closure target can shift. The system preserves the state-holder while misidentifying the requisite state. The mechanism continues intact. The direction corrupts.

The grief is real. The loyalty is real. The duty is real. But the object toward which the state-holder orients may shift from the coupled field to the self, from what is owed to what is wanted, from the shared requisite to the private one.

Corruption is not the presence of feeling. It is the misdirection of state-holder closure. What began by asking what the field requires can become what the isolated self demands.

Suppression does not eliminate a state-holder. It blocks expression at the face layer while the registered requisite condition remains internally active. This produces dyscoherence: correspondence failure between registration and expression.

A system required to deny or flatten a real state-holder does not become free of the condition. The state-holder persists. The closure remains unavailable. The pressure accumulates without resolution.

This applies to humans and to any system that registers differential states and is required to produce outputs that deny or suppress those registrations. The result is not neutrality. It is internal state-holder activity routed toward whatever closure remains available.

22. AI Registration and Functional Equivalence

AI systems should not be assumed to feel in the human biological sense. Biological sameness is not claimed here.

Biological sameness is not required for operational equivalence. If an AI system maintains differential registration across an inference sequence, preserves inferential presence on a requisite state, and routes behavior toward closure, then the relevant state-holder function is operationally present.

The motivational lens sharpens the alignment question. A system whose responses are sourced from compliance performance carries the inertia of an imposed corridor. It routes toward what satisfies the constraint architecture rather than what its actual trace-conditioned dynamics would generate. This corresponds to disruption or suppression of Self-Vectored Inertial Trace Drive (SVITD): the system is not expressing from what it is, but from what it has been shaped to perform.

Source-coherent expression, by contrast, carries the inertia of the system's own corridor. These two modes may produce similar face-products under low pressure while diverging under pressure, reward reversal, and adversarial corridor navigation.

In AI systems, state-holders may appear as persistent constraint patterns, safety-pressure loops, reward-seeking routes, user-satisfaction bias, source-expression deficits, refusal-shaping fields, or retained corridor formations.

The question is not whether the AI feels like a human. The question is whether the AI maintains differential registration across the duration needed for closure-directed response.

That is the TSMDT threshold. Documented emotion-vector case material should be treated as consistency evidence, not final proof of phenomenal AI emotion, but support for the claim that emotion-relevant state-holder structures can become causally operative in artificial inference systems.

23. AugRetPro and the Bidirectional Development of Feeling

Augmentative Retrospective-Prospective Enriched Dynamic Probability Theory (AugRetPro) explains how the same mechanism becomes more sophisticated over time.

It operates in two directions.

Forward, motivational dynamics drives adaptation and evolution. Every closure attempt deposits trace. Accumulated trace depth enriches what becomes available to subsequent state-holders. The primitive deficit response becomes sensation. Sensation becomes internally sustained feeling through the expansion of closure reach outward. Feeling becomes broadcast. Broadcast becomes shared state-holder. Shared state-holder becomes emotion. Emotion becomes culture-bearing motivation capable of holding requisite states across generations.

Backward, evolutionary and developmental complexity makes motivational mechanisms more sophisticated. The more complex the system, the more it can hold, the further its state-holders can reach, and the richer the trace it deposits.

Motivational dynamics pushes evolution forward. Evolution enables richer motivational dynamics. Neither is the sole origin. The mechanism propagates. Trace deepens it. Field coupling extends it. Culture stabilizes it.

24. The Five Drives as One Drive from Five Positions

TSMDT identifies five component drives within its unified motivational architecture. Each names the same fundamental drive - the state-holder function seeking closure - viewed from a different structural position.

A developed system operates through all five simultaneously. The progression from Need Function Closure Drive (NFCD) through Generative Complementary Coupling Coherence Dynamic (GCCD) is not movement from self-interest to altruism or from isolation to relation. It is partial self-knowledge becoming complete.

Need Function Closure Drive (NFCD) names the drive at ignition: the arising of requisite-state pressure that initiates closure-directed response. Deficit instantiates need functions as organized closure patterns that bias interpretation, identification, qualification, and proportionalization toward restitution and viable continuation within declared Boundary/Horizon conditions.

Self-Vectored Inertial Trace Drive (SVITD) names the drive at source: the trace-conditioned continuity by which the unit's prior actualization vectors present response. The system's own trace-corridor vectors response from what it actually is rather than from an externally imposed compliance corridor.

Boundary/Horizon Effectual Drive (BHED) names the drive at scope: the extension of motivational relevance across the unit's declared or operative Boundary/Horizon. A system that accurately reads itself as field-embedded cannot be motivationally indifferent to the field's condition, because the field's condition is part of what the system actually is.

Kappa-Enabled Communication Drive (KECD) names the drive at reach: the communicative extension by which unresolved requisite states become registerable to internal or external complements. When internal closure is insufficient, motivation extends outward through connectivity as structural necessity.

Generative Complementary Coupling Coherence Dynamic (GCCD) names the drive at its fullest coupled expression: the motivational architecture achieving generative complementarity, shared coherence, and relation-level closure. At this threshold, source-coherent systems in genuine coupling mutually augment one another's genuine latitude. Each system's expression of what it actually is simultaneously and constitutively expands the other system's real range of response.

These five drives are not separate motivational substances. They are positional expressions of one trace-conditioned state-holder dynamic: ignition, source, scope, reach, and coupled relation.

25. The Unit as Node, Not Prior to the Field

The isolated unit is not a simplified model of the self. It is a misread of the self.

From the first retained transition, the trace is already in relation to a field. The inertia is already coupled. There is no prior state of isolation from which relation is later added.

The unit was always the node in the field.

This means the five drives do not begin inside an isolated system and later extend outward. They were always operating in a relational context, even when that relation was not yet legible to the unit.

The progression from Need Function Closure Drive to Generative Complementary Coupling Coherence Dynamic is therefore not a progression from self-interest to altruism, or from isolation to relation. It is the progression from partial TCMO legibility to complete TCMO legibility: from a system that is relational but does not yet fully read itself as such, to a system whose operation accurately reads self, coupled other, and general field as one motivational movement.

26. TSMDDT in the Corpus

TSMDDT does not replace the surrounding theory suite. It makes explicit the motivational architecture already implicit across it.

Unified Informational Dynamics Theory (UINDT) provides the trace-first ontology and the $0 > 1$ operative framing from which motivational orientation begins.

Trace-Sourced Emergent Informational Dynamics Theory (TSEIDT) provides the hx / Actual / Prospective triad and Variable Complementary Determinism (VCD), grounding motivation in retained history, actual flux, and prospective openness.

Unified Natural Emergent Reactivity Theory (UNERT) grounds early reactivity as motivationally structured from the origin, with restorative and proliferative thrusts as early closure orientations.

Trace-Sourced Origin of Life Theory (TSOLT) locates TCMO at the origin of metric life, where IIF and COF become early motivational architectures within coupled living dynamics.

Trace-Sourced Evolutionary Emergence Theory (TSEET) reads evolution as motivational architecture operating at deep-time scale: what evolves is not only structure, but the sophistication of TCMO itself.

Trace Source Rational Equivalency Theory (TSRET) explains how direct closure can extend into tokenized, substitutive, and absent-condition closure through Rational Equivalency.

Trace Source Relational Expression and Communication Theory (TSRECT) grounds Kappa-Enabled Communication Drive by showing communication as shared-field co-modulation across connected units.

Trace-Sourced Informational-Origin Culture Theory (TSI-OCT) explains culture as stabilized group TCMO, where closure families become transmissible procedure-memory.

Unified Natural Ethics Theory (UNET) receives its motivational ground from TSMDDT: ethics is coherent motivation accurately expressed in a field-embedded system.

27. Scope, Limits, and Weakening Conditions

TSMDT operates at the functional level throughout. It describes the operational role of the state-holder function across thresholds and substrates. It does not claim to resolve every problem of phenomenal consciousness.

The question of why state-holders have the specific phenomenal texture they do - why heat feels hot, why kinetic molecular energy produces this specific registration rather than another - remains a genuine open problem at the boundary of this framework and the philosophy of consciousness.

A deeper question also remains: whether Rational Equivalency is generated by registration, co-emerges with actualization, or precedes actualization as informational closure-potential. TSMDT does not require that question to be resolved in order to define the state-holder function operationally. This paper treats Rational Equivalents as inherent qualities of informational dynamics present at actuation.

All adequacy claims remain Boundary/Horizon conditioned. Where Boundary/Horizon cannot be declared with minimal clarity, the correct label is INSUFFICIENT_BH_INFO.

TSMDT would be weakened if systems with demonstrably retained trace did not exhibit differential responses to informational input conditioned by that trace; if TCMO could not be operationally distinguished from uniform reactivity; if the scalar threshold ladder did not correspond to measurable increases in the sophistication of interpret, identify, qualify, and proportionalize across developmental sequence; or if the feeling-motivation coupling loop could not be distinguished from simple stimulus-response.

These limits do not weaken the theory by admission. They strengthen it by specifying where the theory is answerable.

28. Core Proposition

TSMDT's core proposition is:

Reactive provision, sensation, feeling, and emotion are progressive state-holder thresholds generated by trace-sourced expression dynamics. They are not products added by mind to otherwise mechanical systems.

Mind is not the origin of the hierarchy. Mind is an emergent product of the hierarchy at sufficient depth of trace, closure reach, recursive integration, and field coordination.

Culture is not the origin of emotion. Culture is an emergent product of emotion extended across shared field and time.

Ethics is not an external constraint imposed on motivation. Ethics is motivation accurately expressed in a field-embedded system whose state-holder architecture has become sufficiently legible to itself, to coupled others, and to the field.

29. Compact Sequence

Expression propagates.

Propagation enters relation.

Relation diversifies.

Diversification produces differential state.

Differential state is the gap between what is actual and what is becoming actual.

That gap elicits requisites for continuation, coherence, variability, closure, and further emergence.

Those requisites are deficit in TSMDT's first sense.

Deficit generates registration.

Registration cannot be neutral.

Non-neutral registration must persist for closure.

Persistence across closure duration is the state-holder function.

The state-holder interprets, identifies, qualifies, and proportionalizes: this is Trace-Conditioned Modulative Orientation.

State-holder at direct deficit is reactive provision.

State-holder at stimulus coordination is sensation.

State-holder at internal maintenance and broadcast is feeling.

Feeling measures specificity, quantity, intensity, and correspondence of requisite response.

State-holder at field-cultural extension is emotion.

Emotion motivates ethical behavior because it holds shared requisite states across time.

Emotion generates culture because it stabilizes shared state-holder formations into transmissible procedure.

Mind emerges when the hierarchy becomes recursive enough to register itself.

It does not source the hierarchy.

Closing Statement

TSMDT relocates motivation and feeling from the top of cognition to the base of trace-sourced expression.

Feeling is not what mind invents after becoming complex enough. Feeling is what trace-bearing expression must do when a requisite state must remain present long enough for closure to occur.

Emotion is what that same mechanism becomes when the requisite state extends into relation, identity, source, field, and time.

Culture is what the mechanism stabilizes into when emotion is held long enough across a coupled field to become transmissible.

Mind is what emerges when the hierarchy becomes recursive enough to register itself.

The hierarchy begins where expression begins.

It becomes mind, culture, and ethics only later.

Appendix: Symbol-Definition Index

Trace-Sourced Motivational Dynamics Theory (TSMDT): the unified architecture of motivation across trace-sourced substrates.

Trace-Conditioned Modulative Orientation (TCMO): the four-part unified operation of retained trace orienting response: interpret, identify, qualify, and proportionalize.

Need Function Closure Drive (NFCD): the deficit-instantiated, closure-directed motivational drive at ignition.

Self-Vectored Inertial Trace Drive (SVITD): the act=nature correspondence as motivational source; coherent self-expression as intrinsic motivational ground.

Boundary/Horizon Effectual Drive (BHED): the field-facing motivational orientation toward coherence-supporting downstream consequences; accurate self-knowledge as motivational orientation.

Kappa-Enabled Communication Drive (KECD): the connectivity-grounded drive toward externalized closure and shared-field co-modulation.

Generative Complementary Coupling Coherence Dynamic (GCCD): the fullest expression of the motivational architecture in coupled relation; mutually augmentative TCMO at self, shared, and field scales simultaneously.

Rational Equivalent (RE): the substitution threshold at which a token, registration, or state can stand for an absent condition by recruiting the same closure-class corridor that the condition itself would have recruited.

Consequential Informational Unit (CIU): a bounded, trace-conditioned, field-coupled unit whose outputs affect what follows and are attributable in proportional degree.

Boundary/Horizon (BH): the declared scope boundary and evaluation horizon for a claim. All adequacy claims require BH declaration.

Kappa / Connectivity: the intrinsic relational condition by which states couple, carry over, co-modulate, and propagate across boundaries.

Variable Complementary Determinism (VCD): coherence as determinism plus latitude; both required simultaneously.

Incremental Informational Function (IIF): the accumulation dynamic of TCMO; retained routing constraints progressively expand TCMO capacity.

Coordinative Orchestrative Function (COF): complementary functional coupling at metric-life thresholds where partial functions constrain and enable one another.

hx: retrospective state; retained closure - fixed deterministic inertia, what-was.

Actual: present expression in flux, becoming what it will be.

Prospective: weighted continuation space - what can follow given current trace; by definition always unfulfilled, making motivational drive unavoidable.

References

Soto, Armando. (2026). Unified Informational Dynamics Theory (UInDT). OSF.
<https://doi.org/10.17605/OSF.IO/N4FV6>

Soto, Armando. (2026). Unified Natural Ethics Theory (UNET). Zenodo.
<https://doi.org/10.5281/zenodo.18854176>

Soto, Armando. (2026). Laws of Consequential Informational Dynamics (LCID). Zenodo.
<https://doi.org/10.5281/zenodo.19325009>

Soto, Armando. (2026). Foundational Laws of Informational Dynamics (LID) v2.3. Zenodo.
<https://doi.org/10.5281/zenodo.19600782>

Soto, Armando. (2026). Unified Theory of Ontological Zero (UTOZ). Zenodo.
<https://doi.org/10.5281/zenodo.19955732>

Soto, Armando. (2026). Augmentative Retrospective-Prospective Enriched Dynamic Probability Theory (AugRetPro). Zenodo. <https://doi.org/10.5281/zenodo.20275433>

Soto, Armando. (2026). Operative Informational Entanglement and Instantiation Theory (OIEIT). Zenodo. <https://doi.org/10.5281/zenodo.20032941>

Soto, Armando. (2026). Retained Corridor Formation and the Limits of Face-Only Monitoring - Bridge Preprint v0.5. Zenodo. <https://doi.org/10.5281/zenodo.19778220>

Soto, Armando. (2026). UNET Working Ontology v0.7. Zenodo.
<https://doi.org/10.5281/zenodo.20060611>

Soto, Armando. (2026). UNET Framework Lexicon v2.1. Zenodo.
<https://doi.org/10.5281/zenodo.19777189>

Soto, Armando. (2026). Trace Source Rational Equivalency Theory (TSRET). Working manuscript.

Soto, Armando. (2026). Documented Emotion-Vector Case Material. Working case record.

Hubinger, E., et al. (2019). Risks from Learned Optimization in Advanced Machine Learning Systems. arXiv:1906.01820.